

1 Split, apply, combine



```
orders.groupby("customer_id").agg(
    frequency=("order_id", "count"),
    monetary=("amount", "sum"),
    last_order=("order_date", "max"),
)
```

Crux. `groupby` is the Week 1 accumulator with the splitting and the combining done for you, leaving only the part that was ever about your question.

2 Name every aggregation

Form	What it costs
<code>.sum()</code>	Shorter to type, and six months later nobody can tell whether a column was chosen or inherited
<code>.agg(name=(col, fn))</code>	Longer to type, and every column says where it came from

State a reference date for recency. Counting from today makes the table change meaning every time it is rebuilt.

Crux. The short form sums every numeric column, including ones nobody meant to add up, and names them after the inputs.

3 merge is join, with the same danger

```
table.merge(feed, on="customer_id",
            how="left", validate="one_to_one")
```

```
pandas.errors.MergeError: Merge keys are not
unique in right dataset; not a one-to-one merge
```

Crux. A merge fans out exactly like a join, and pandas will refuse if you tell it what you believe.

4 The four validate values

Value	Promises
<code>one_to_one</code>	Keys unique on both sides
<code>one_to_many</code>	Keys unique on the left
<code>many_to_one</code>	Keys unique on the right
<code>many_to_many</code>	Nothing, and says so out loud

Fixing a failure is a decision, not a keyword: drop duplicates and you pick a row arbitrarily, aggregate first and you keep both facts, ask the feed's owner and you fix it upstream.

Crux. Each value is a claim about the data, so choosing one forces you to say what you believe before the data disagrees.

5 Reshape changes the question

```
wide = long.pivot_table(index="customer_id",
                        columns="month", values="spend", aggfunc="sum")
back = wide.reset_index().melt(id_vars="customer_id")
```

Shape	One row is	Easy question
Long	A customer-month	How did this move
Wide	A customer	How do these compare

Read the row labels aloud before reading any value. "Each row is one order" is how a wrong index gives itself away.

Crux. `pivot_table` widens and `melt` lengthens, and nothing is added or removed by either.

6 Which tool owns which number

Who owns this number, for how long, and who has to be able to read it.

Tool	Owns	Never
Warehouse	Numbers Finance acts on	Exploration
pandas	The analyst's iteration	The source of truth
Plain Python	What you explain line by line	Anything at scale

A notebook has no audit trail a controller can read, and a cell edited at 4 pm looks identical to one nobody touched.

Crux. Three questions pick the tool nearly every time, and none of them is about syntax.